



Modelling Financial Markets

Seminar 10: Market Microstructure — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EKOP: Easley–Kiefer–O'Hara–Paperman (model of informed trading)
- ▣ ETF: Exchange-Traded Fund
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ MA: Moving Average
- ▣ OLS: Ordinary Least Squares
- ▣ PIN: Probability of INformed trading
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VECM: Vector Error-Correction Model
- ▣ VIX: CBOE Volatility IndeX

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the moments and the standard error of the Roll estimator, and its bias under autocorrelated order flow
- ▣ Derive the Glosten–Milgrom spread for any prior and the Kyle equilibrium; read λ as a regression slope
- ▣ Solve Almgren–Chriss by Euler–Lagrange; write a numerically stable PIN likelihood
- ▣ Do inference on real data: bootstrap, HAC, Monte Carlo under the null, break tests, IV (Instrumental Variable) estimation, VECM price discovery, predictive-regression bias

- ▣ Background: Lecture 10; textbooks Hasbrouck (2007), Bouchaud et al. (2018), Franke, Härdle & Hafner (2019)

Structure

- ▣ **Part A:** ten derivations (A1–A10)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), the critique of an AI answer (C2), and a project seed from a landmark paper (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Units: relative spreads and price moves in basis points ($1 \text{ bp} = 0.01\%$); traded value in USD
- Intraday data have strong dependence within a day: resample whole days, not single bars

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Roll moments and bias), A3 (Glosten–Milgrom for any prior), A5 (deriving Kyle), A7 (Almgren–Chriss by Euler–Lagrange), A9 (the PIN likelihood)
- Part B: B1 (the intraday U-shape), B2 (spread estimators at two frequencies, Monte Carlo of Roll), B4 (illiquidity and the VIX, HAC, break test), B6 (volume and price moves, IV), B8 (price discovery, VECM)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 (model: A5), A8 (model: A7), A10 (model: A9)
- Part B: B3 (spreads across 25 assets; model: B2), B5 (Amihud on three markets; model: B2, B4), B7 (the Bitcoin clock; model: B1)
- Part C: C1, open question and project seed; C2, critique of an AI answer; C3, the price of the speed race in SPY (case study)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter10_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- SPY 5-minute bars, regular session, October 2020 – September 2026; Bitcoin 5-minute bars, September 2024 – September 2026
- Daily open, high, low, close and volume for 7 US stocks, 11 BVB stocks and 7 crypto-assets; VIX; BET-TR (BET with dividends reinvested)
- BVB traded values converted from lei to USD at the BNR reference rate of each day

Formulas Used Today

- Roll bias and precision [Harris \(1990\)](#): $\mathbb{E}[\hat{\gamma}_1] \approx \gamma_1 - (\gamma_0 + 2\gamma_1)/T$, $T\text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$ (Gaussian MA(1))
- Roll [Roll \(1984\)](#): $\text{Var}(\Delta p) = \sigma^2 + 2c^2$, $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$, $s = 2c$
- Glosten–Milgrom [Glosten & Milgrom \(1985\)](#): $a = V_L + (V_H - V_L)P(V_H \mid \text{buy})$, $b = V_L + (V_H - V_L)P(V_H \mid \text{sell})$, Bayes' rule
- Kyle [Kyle \(1985\)](#): $\beta = \sigma_u/\sigma_v$, $\lambda = \sigma_v/(2\sigma_u)$, expected profit $\sigma_v\sigma_u/2$; PIN [Easley et al. \(1996\)](#): $\alpha\mu/(\alpha\mu + \varepsilon_b + \varepsilon_s)$
- Almgren–Chriss [Almgren & Chriss \(2001\)](#): $x_k = X \sinh(\kappa(T - t_k))/\sinh(\kappa T)$, $\cosh(\kappa T) = 1 + \tilde{\kappa}^2 \tau^2/2$, $\tilde{\kappa}^2 = \lambda\sigma^2/\tilde{\eta}$, $\tilde{\eta} = \eta - \gamma\tau/2$
- Amihud [Amihud \(2002\)](#): $\text{ILLIQ} = \frac{1}{D} \sum_d |r_d|/\text{DVOL}_d$

A1: The Roll Model: Moments, Estimator, Bias [Solved]

Task

□ Context and assumptions

- ▶ Efficient price: a random walk $m_t = m_{t-1} + \varepsilon_t$, with ε_t i.i.d., mean 0, variance σ^2 Roll (1984).
- ▶ Trade price: $p_t = m_t + c q_t$, where c is the half-spread and $q_t = +1$ (buy) or -1 (sell), independent of ε .
- ▶ Numbers for (b): a BVB stock trading around 20 lei, with daily price changes $\text{Var}(\Delta p) = 0.0520$ and $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -0.0081$ (lei²).

□ Sub-tasks

- ▶ (a) Assume the q_t are i.i.d. with $P(q_t = 1) = 1/2$. Derive $\text{Var}(\Delta p_t)$ and $\text{Cov}(\Delta p_t, \Delta p_{t-1})$ from the definition of Δp_t .
- ▶ (b) With the numbers above, compute c , the spread $s = 2c$ in lei and in % of the price, the efficient variance σ^2 , the share of $\text{Var}(\Delta p)$ due to the bounce, and the first-order autocorrelation ρ_1 .
- ▶ (c) Now let the trade signs be a Markov chain with $\text{Corr}(q_t, q_{t-k}) = \rho^k$ and no information content. Derive $\text{Cov}(\Delta p_t, \Delta p_{t-1})$ and the limits of the Roll half-spread estimator $\sqrt{-\hat{\gamma}_1}$ and spread estimator $2\sqrt{-\hat{\gamma}_1}$ for $c = 0.090$, $\rho = 0.3$.

□ Report and interpret

- ▶ The two moments, the five numbers of (b) and the two limits of (c).
- ▶ Interpretation: does a positive ρ make the Roll estimator too high or too low, and why?

A1: solution [Solved]

Solution (1/2)

- Step 1. Differencing gives $\Delta p_t = \varepsilon_t + c(q_t - q_{t-1})$: the efficient innovation plus the change in the side of the trade.
- Step 2. Since ε and q are independent, $\text{Var}(\Delta p_t) = \sigma^2 + c^2 \text{Var}(q_t - q_{t-1}) = \sigma^2 + c^2(1 + 1) = \sigma^2 + 2c^2$.
- Step 3. $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = c^2 \text{Cov}(q_t - q_{t-1}, q_{t-1} - q_{t-2})$; the only common term is q_{t-1} , with sign $-$, so the covariance is $-c^2 \text{Var}(q_{t-1}) = -c^2$.
- Step 4 (b). $c = \sqrt{0.0081} = 0.090$ lei, $s = 2c = 0.18$ lei, i.e. $0.18/20 = 0.9\%$ of the price.
- Step 5 (b). $\sigma^2 = 0.0520 - 2c^2 = 0.0520 - 0.0162 = 0.0358$; bounce share $2c^2/\text{Var}(\Delta p) = 31.2\%$;
 $\rho_1 = -0.0081/0.0520 = -0.156$.

A1: solution [Solved]

Solution (2/2)

- Step 6 (c). With $\text{Cov}(q_t, q_{t-k}) = \rho^k$: $\text{Cov}(q_t - q_{t-1}, q_{t-1} - q_{t-2}) = \rho - \rho^2 - 1 + \rho = -(1 - \rho)^2$.
- Step 7 (c). Hence $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2(1 - \rho)^2 = -0.0081 \cdot 0.49 = -0.0040$.
- Step 8 (c). The Roll half-spread estimator converges to $\sqrt{c^2(1 - \rho)^2} = c(1 - \rho) = 0.063$ lei instead of 0.090, and the spread estimator to 0.126 instead of 0.18 lei: both 30% too low.
- Interpretation: almost a third of the daily variance is bid-ask noise, so volatility from trade prices overstates efficient volatility.
- Interpretation: persistent order flow (orders split into several trades) makes reversals rarer, hides part of the bounce and biases the Roll spread downwards.

A2: The Standard Error of the Roll Estimator [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Same population moments as A1(b): $\gamma_0 = \text{Var}(\Delta p) = 0.0520$, $\gamma_1 = \text{Cov}(\Delta p_t, \Delta p_{t-1}) = -0.0081$, so $c = 0.090$ and $s = 0.18$ lei [Harris \(1990\)](#).
- ▶ The estimator is $\hat{s} = 2\sqrt{-\hat{\gamma}_1}$, with $\hat{\gamma}_1$ the sample autocovariance from T price changes.
- ▶ Use the Gaussian approximation (Bartlett's formula for an MA(1)): $T \text{Var}(\hat{\gamma}_1) \rightarrow \gamma_0^2 + 3\gamma_1^2$.

□ Sub-tasks

- ▶ (a) Apply the delta method to $g(\gamma_1) = 2\sqrt{-\gamma_1}$ and write $\text{se}(\hat{s})$ as a function of γ_0 , γ_1 and T .
- ▶ (b) Evaluate $\text{se}(\hat{\gamma}_1)$, $\text{se}(\hat{s})$ and $\text{se}(\hat{s})/s$ for $T = 78$ (one day of 5-minute bars) and $T = 250$ (one year of daily data).
- ▶ (c) Approximate $P(\hat{\gamma}_1 > 0)$ for both T with the Normal distribution, and explain why this approximation is only rough here.

□ Report and interpret

- ▶ A table with T , $\text{se}(\hat{\gamma}_1)$, $\text{se}(\hat{s})$, relative error and $P(\hat{\gamma}_1 > 0)$.
- ▶ Interpretation: how often would a true Roll market show a positive covariance on a single day?

□ Model: A1

A3: Glosten–Milgrom for Any Prior [Solved]

Task

□ Context and assumptions

- ▶ The asset is worth V_L or V_H , $\Delta V = V_H - V_L$, with prior $P(V = V_H) = \theta$ [Glosten & Milgrom \(1985\)](#).
- ▶ One trader arrives per period: informed with probability μ (buys if $V = V_H$, sells otherwise), uninformed otherwise (buys or sells with probability $1/2$).
- ▶ A competitive risk-neutral market maker sets $a = \mathbb{E}[V \mid \text{buy}]$ and $b = \mathbb{E}[V \mid \text{sell}]$. Numbers for (c): $V_L = 90$, $V_H = 110$, $\mu = 0.3$.

□ Sub-tasks

- ▶ (a) Use Bayes' rule to derive $P(V_H \mid \text{buy})$ and $P(V_H \mid \text{sell})$, then the spread $a - b$ as a function of θ , μ and ΔV ; show that it is largest at $\theta = 1/2$.
- ▶ (b) Show that the expected price of the next trade equals $\mathbb{E}[V \mid \mathcal{F}_t]$ and that this expectation is a martingale.
- ▶ (c) Compute the ask, the bid and the spread at $\theta = 0.5$, the new quotes after one buy, and the spread at $\theta = 0.7$ from your formula.

□ Report and interpret

- ▶ The spread formula, the martingale argument and the quotes of (c).
- ▶ Interpretation: why is the spread largest when the market maker is most uncertain?

A3: solution [Solved]

Solution (1/2)

- Step 1. $P(\text{buy} \mid V_H) = \mu + (1 - \mu)/2 = (1 + \mu)/2$ and $P(\text{buy} \mid V_L) = (1 - \mu)/2$.
- Step 2. Bayes, with $D = 2\theta - 1$: $P(V_H \mid \text{buy}) = \theta(1 + \mu)/(1 + \mu D)$ and $P(V_H \mid \text{sell}) = \theta(1 - \mu)/(1 - \mu D)$.
- Step 3. $a - b = \Delta V [P(V_H \mid \text{buy}) - P(V_H \mid \text{sell})] = 4\mu\theta(1 - \theta) \Delta V / (1 - \mu^2(2\theta - 1)^2)$.
- Step 4. With $x = 2\theta - 1$, $4\theta(1 - \theta) = 1 - x^2$ and the spread is $\mu\Delta V(1 - x^2)/(1 - \mu^2x^2) \leq \mu\Delta V$, with equality only at $x = 0$, i.e. $\theta = 1/2$.
- Step 5 (b). $P(\text{buy})a + P(\text{sell})b = \mathbb{E}[V \mid \mathcal{F}_t]$ by the law of total expectation; $\mathbb{E}[\mathbb{E}[V \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \mathbb{E}[V \mid \mathcal{F}_t]$ by iterated expectations, so beliefs and expected trade prices are martingales.

A3: solution [Solved]

Solution (2/2)

- Step 6 (c). At $\theta = 0.5$: $P(V_H \mid \text{buy}) = 0.65 \cdot 0.5 / (0.65 \cdot 0.5 + 0.35 \cdot 0.5) = 0.6500$, so $a = 90 + 20 \cdot 0.6500 = 103.00$; symmetrically $b = 97.00$; spread $6.00 = \mu \Delta V$.
- Step 7 (c). After one buy, $\theta = 0.6500$ and $P(V_H \mid \text{buy}) = 0.7752$: $a = 105.50$, $b = 100.00$, spread 5.50 .
- Step 8 (c). At $\theta = 0.7$: $4 \cdot 0.3 \cdot 0.21 \cdot 20 / (1 - 0.09 \cdot 0.16) = 5.11$, the same as the quotes computed directly.
- Interpretation: at $\theta = 1/2$ a single trade moves beliefs most, so adverse selection costs most; once beliefs lean one way, less remains to be learned and the spread narrows.

A4: How Fast Does the Market Maker Learn? [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Glosten–Milgrom as in A3 with $V_L = 90$, $V_H = 110$ and $\theta_0 = 1/2$.
- ▶ The true value is V_H and, to isolate the speed of learning, assume every trade is a buy.

□ Sub-tasks

- ▶ (a) Show that each buy multiplies the odds $\theta/(1 - \theta)$ by $(1 + \mu)/(1 - \mu)$, and that the bid after k buys equals $V_L + \Delta V \theta_{k-1}$.
- ▶ (b) Derive the smallest number of buys k after which the bid exceeds 108, and evaluate it for $\mu = 0.1$, 0.3 and 0.5.
- ▶ (c) Compare with the simulated learning paths of the lecture (Section 6), where uninformed traders also sell.

□ Report and interpret

- ▶ The formula for k and the three values.
- ▶ Interpretation: how does the speed of price discovery depend on the share of informed traders?

□ Model: A3

A5: Deriving the Kyle Equilibrium [Solved]

Task

▣ Context and assumptions

- ▶ One auction. The value is $v \sim N(p_0, \sigma_v^2)$; one insider observes v and submits a market order x [Kyle \(1985\)](#).
- ▶ Noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v ; the market maker sees only $y = x + u$ and sets $p = \mathbb{E}[v \mid y]$.
- ▶ Numbers: $\sigma_v = 2$ USD, $\sigma_u = 10\,000$ shares; a second session with $\sigma_u = 20\,000$ shares.

▣ Sub-tasks

- ▶ (a) Conjecture $p = p_0 + \lambda y$. Derive the insider's optimal x from the first-order condition and write β in $x = \beta(v - p_0)$.
- ▶ (b) Given $x = \beta(v - p_0)$, derive λ by the projection theorem, solve the two equations for β and λ , and compute the expected insider profit.
- ▶ (c) Compute β , λ , the profit and the price change for $y = 15\,000$; then compare λ and the profit in the second session.

▣ Report and interpret

- ▶ The equilibrium formulas and the numbers of (c).
- ▶ Interpretation: why does more noise trading help the insider?

A5: solution [Solved]

Solution (1/2)

- Step 1. Given v , the expected profit is $\mathbb{E}[(v - p_0 - \lambda x - \lambda u)x \mid v] = (v - p_0)x - \lambda x^2$ (since $\mathbb{E}[u] = 0$).
- Step 2. First-order condition: $(v - p_0) - 2\lambda x = 0$, so $x = (v - p_0)/(2\lambda)$ and $\beta = 1/(2\lambda)$.
- Step 3. (v, y) is jointly Normal, so $\mathbb{E}[v \mid y] = p_0 + \frac{\text{Cov}(v, y)}{\text{Var}(y)}y$ with $\lambda = \frac{\beta \sigma_v^2}{\beta^2 \sigma_v^2 + \sigma_u^2}$.
- Step 4. Substituting $2\lambda\beta = 1$: $\beta^2 \sigma_v^2 + \sigma_u^2 = 2\beta^2 \sigma_v^2$, so $\beta = \sigma_u/\sigma_v$ and $\lambda = \sigma_v/(2\sigma_u)$.
- Step 5. Expected profit $\mathbb{E}[(v - p_0)x - \lambda x^2] = \beta \sigma_v^2 - \lambda \beta^2 \sigma_v^2 = \sigma_v \sigma_u/2$.

A5: solution [Solved]

Solution (2/2)

- ▣ Step 6 (c). $\beta = 10\,000/2 = 5\,000$ shares per USD of mispricing; $\lambda = 2/20\,000 = 0.0001$ USD per share.
- ▣ Step 7 (c). Expected profit $2 \cdot 10\,000/2 = 10\,000$ USD; for $y = 15\,000$, $\Delta p = 0.0001 \cdot 15\,000 = 1.50$ USD.
- ▣ Step 8 (c). With $\sigma_u = 20\,000$: $\lambda = 0.00005$ (half), $\beta = 10\,000$ and profit $20\,000$ USD (double).
- ▣ Interpretation: more noise makes the order flow less informative, so the market maker moves prices less per share; the insider trades more aggressively and earns more, while half of the information still enters the price.

A6: Kyle's Lambda as a Regression Slope [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ The Kyle economy of A5 ($\sigma_v = 2$ USD, $\sigma_u = 10\,000$ shares) with $p_0 = 50$ USD, repeated over many independent auctions.
- ▶ The notebook simulates 200 000 auctions to check the population results numerically.

□ Sub-tasks

- ▶ (a) Show that the population OLS slope of $p - p_0$ on y , and of $v - p_0$ on y , both equal λ ; give $\text{Var}(y)$.
- ▶ (b) Derive what the Amihud-type ratio $|r|/(p_0|y|)$, with $r = (p - p_0)/p_0$, estimates, and express it in bp per USD 1 million.
- ▶ (c) Suppose the denominator is the gross customer-order proxy $|x| + |u|$ (insider and aggregate noise order counted separately, a convention, not the executed volume of the auction) instead of the net flow $|y|$: derive the direction of the bias and quantify it by simulation.

□ Report and interpret

- ▶ The three results and the simulated ratios.
- ▶ Interpretation: what does the Amihud ratio measure in a Kyle world, and why is it biased?

□ Model: A5

A7: Almgren–Chriss by Euler–Lagrange [Solved]

Task

- Context and assumptions
 - ▶ Sell $X = 10^6$ shares within $T = 5$ days; parameters of the numerical example of [Almgren & Chriss \(2001\)](#): $\sigma = 0.95$ USD per day^{1/2}, temporary impact $\eta = 2.5 \cdot 10^{-6}$, permanent impact $\gamma = 2.5 \cdot 10^{-7}$, risk aversion $\lambda = 2 \cdot 10^{-6}$.
 - ▶ The trader minimises $\mathbb{E}[C] + \lambda \text{Var}[C]$, where C is the implementation cost.
- Sub-tasks
 - ▶ (a) In continuous time, minimise $\int_0^T (\eta \dot{x}^2 + \lambda \sigma^2 x^2) dt$ subject to $x(0) = X$, $x(T) = 0$: write the Euler–Lagrange equation, solve it and give κ_c .
 - ▶ (b) In discrete time with $N = 5$ daily steps ($\tau = 1$), compute $\tilde{\eta} = \eta - \gamma\tau/2$, $\tilde{\kappa}^2 = \lambda\sigma^2/\tilde{\eta}$, κ from $\cosh(\kappa\tau) = 1 + \tilde{\kappa}^2\tau^2/2$, and the holdings x_k .
 - ▶ (c) Compute $\mathbb{E}[C]$ and the standard deviation of C , and compare them with selling evenly ($\lambda = 0$).
- Report and interpret
 - ▶ $x(t)$, κ_c and κ , the schedule, and the cost–risk comparison.
 - ▶ Interpretation: what does the trader buy with the extra expected cost?

A7: solution [Solved]

Solution (1/2)

- Step 1. With $F(x, \dot{x}) = \eta \dot{x}^2 + \lambda \sigma^2 x^2$, Euler–Lagrange $\frac{d}{dt} \partial F / \partial \dot{x} = \partial F / \partial x$ gives $2\eta \ddot{x} = 2\lambda \sigma^2 x$, i.e. $\ddot{x} = \kappa_c^2 x$.
- Step 2. $\kappa_c^2 = \lambda \sigma^2 / \eta = 2 \cdot 10^{-6} \cdot 0.9025 / (2.5 \cdot 10^{-6}) = 0.722$, $\kappa_c = 0.8497$.
- Step 3. The general solution $Ae^{\kappa_c t} + Be^{-\kappa_c t}$ with $x(0) = X$, $x(T) = 0$ gives $x(t) = X \sinh(\kappa_c(T - t)) / \sinh(\kappa_c T)$; e.g. $x(1) = 427\,151$, $x(2) = 181\,712$.
- Step 4 (b). $\tilde{\eta} = 2.5 \cdot 10^{-6} - 1.25 \cdot 10^{-7} = 2.375 \cdot 10^{-6}$; $\tilde{\kappa}^2 = 2 \cdot 10^{-6} \cdot 0.9025 / (2.375 \cdot 10^{-6}) = 0.76$.
- Step 5 (b). $\cosh \kappa = 1 + 0.76/2 = 1.38$, so $\kappa = 0.8463$ and $\sinh(5\kappa) = 34.40$.

A7: solution [Solved]

Solution (2/2)

- ▣ Step 6 (b). $x_k = X \sinh(\kappa(5 - k)) / \sinh(5\kappa)$: $x = (1\,000\,000; 428\,599; 182\,933; 76\,296; 27\,643; 0)$.
- ▣ Step 7 (b). Daily sales $n_k = x_{k-1} - x_k$: $(571\,401; 245\,666; 106\,637; 48\,652; 27\,643)$, i.e. 57% on day 1.
- ▣ Step 8 (c). $\mathbb{E}[C] = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau} \sum n_k^2 = 1.078$ million USD and $\text{sd}[C] = \sigma \sqrt{\tau \sum x_k^2} = 0.449$ million USD.
- ▣ Step 9 (c). Selling evenly gives $\mathbb{E}[C] = 0.600$ and $\text{sd}[C] = 1.041$ million USD.
- ▣ Interpretation: front-loading pays about 0.48 million USD more in expected impact to cut the standard deviation of the cost by more than half; the continuous and discrete κ happen to be close for these parameters ($\kappa\tau \approx 0.85$ is not small) because the discretisation and the adjusted impact $\tilde{\eta}$ partly offset.

A8: A More Risk-Averse Seller [Proposed]

Solution discussed in class

Task

- Context and assumptions
 - ▶ Same order and parameters as A7, but a ten times higher risk aversion, $\lambda = 2 \cdot 10^{-5}$.
- Sub-tasks
 - ▶ (a) Compute $\tilde{\kappa}^2$, the discrete κ , the continuous κ_c and the holdings x_k for $N = 5$.
 - ▶ (b) Explain why the discrete and continuous κ now differ much more than in A7.
 - ▶ (c) Compute $\mathbb{E}[C]$ and $\text{sd}[C]$ and the ratios to A7.
- Report and interpret
 - ▶ The schedule, the two κ and the cost–risk ratios.
 - ▶ Interpretation: when would a trader choose this schedule?
- **Model:** A7

A9: The PIN Likelihood, Step by Step [Solved]

Task

□ Context and assumptions

- ▶ EKOP model [Easley et al. \(1996\)](#): each day an information event occurs with probability α ; it is bad news with probability δ .
- ▶ Given the day's information state, daily buy and sell counts B and S are independent Poisson: uninformed rates $\varepsilon_b, \varepsilon_s$; informed traders add μ to the buys on good-news days and to the sells on bad-news days.
- ▶ Numbers: $\alpha = 0.3$, $\delta = 0.5$, $\mu = 400$, $\varepsilon_b = \varepsilon_s = 1\,000$; one day with $B = 1\,450$ and $S = 1\,000$.

□ Sub-tasks

- ▶ (a) Write the likelihood of one day as a three-component Poisson mixture, and the probability of informed trading PIN.
- ▶ (b) Compute PIN and explain why evaluating the likelihood directly in double precision fails for this day.
- ▶ (c) Factor out the common terms, derive a numerically stable log-likelihood, evaluate it with log-sum-exp, and compute the posterior probability of a good-news day.

□ Report and interpret

- ▶ PIN, $\ln L$ and the posterior probability.
- ▶ Interpretation: why are days classified almost surely when counts are large?

A9: solution [Solved]

Solution (1/2)

- Step 1. With $\pi_n(r) = e^{-r} r^n / n!$: $L = (1 - \alpha)\pi_B(\varepsilon_b)\pi_S(\varepsilon_s) + \alpha\delta\pi_B(\varepsilon_b)\pi_S(\mu + \varepsilon_s) + \alpha(1 - \delta)\pi_B(\mu + \varepsilon_b)\pi_S(\varepsilon_s)$.
- Step 2 (b). $\text{PIN} = \alpha\mu / (\alpha\mu + \varepsilon_b + \varepsilon_s) = 120 / 2\,120 = 0.0566$.
- Step 3 (b). In double precision e^{-1000} underflows to 0 and 1000^{1450} overflows to ∞ , so $0 \cdot \infty$ returns NaN and the optimiser fails or stops at a boundary [Lin & Ke \(2011\)](#).

A9: solution [Solved]

Solution (2/2)

- Step 4 (c). Factor out $K = e^{-\varepsilon_b - \varepsilon_s} (\mu + \varepsilon_b)^B (\mu + \varepsilon_s)^S / (B! S!)$ and set $x_b = \varepsilon_b / (\mu + \varepsilon_b)$, $x_s = \varepsilon_s / (\mu + \varepsilon_s)$.
- Step 5 (c). Then $L = K [(1 - \alpha)x_b^B x_s^S + \alpha\delta e^{-\mu} x_b^B + \alpha(1 - \delta)e^{-\mu} x_s^S]$; $\ln K$ uses $\ln B!$ from the log-gamma function.
- Step 6 (c). Take the log of each bracket term and combine them with log-sum-exp: $\ln L = -11.71$.
- Step 7 (c). The posterior probability of a state is its term's share of the bracket: good news 1.0000.
- Interpretation: with 450 extra buys against a Poisson standard deviation of about 32, the day is classified almost surely; this is why PIN is identified from the share of such days, not from single days.

A10: Reading Days with the PIN Model [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Same EKOP parameters as A9: $\alpha = 0.3$, $\delta = 0.5$, $\mu = 400$, $\varepsilon_b = \varepsilon_s = 1\,000$.

□ Sub-tasks

- ▶ (a) For a day with $B = 1\,000$ and $S = 1\,420$, compute the stable log-likelihood and the posterior probability of bad news.
- ▶ (b) For a quiet day with $B = S = 1\,000$, compute the posterior probability of no news.
- ▶ (c) Compute PIN when μ doubles to 800, and explain why PIN remains hard to estimate for very liquid stocks even with the stable likelihood.

□ Report and interpret

- ▶ The two posterior probabilities, $\ln L$ and the new PIN.
- ▶ Interpretation: can an order imbalance always be read as private information?

□ Model: A9

B1: The Intraday U-Shape of SPY [Solved]

Task

▣ Context and assumptions

- ▶ SPY 5-minute bars (data/market/intraday/SPY.US_5m.csv), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Returns are log changes of the close within a day; the first bar's return runs from the day's open to the first bar's close. $|r|$ is in basis points.

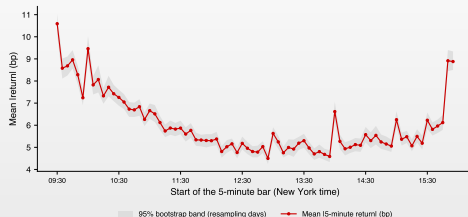
▣ Sub-tasks

- ▶ (a) Compute the mean $|r|$ of the first bar (09:30) and of the midday bars (11:30–14:00) over all days, and their ratio.
- ▶ (b) Compute the average share of daily volume traded in the last six bars (30 minutes).
- ▶ (c) Build 95% percentile bootstrap intervals for both statistics by resampling whole days with replacement (1 000 resamples, seed 42) [Efron \(1979\)](#).

▣ Report and interpret

- ▶ The two point estimates with intervals and the chart of mean $|r|$ by bar with its bootstrap band.
- ▶ Interpretation: why resample days rather than bars; is the U-shape systematic; what does it imply for a trader who must buy 1 million shares today?

B1: solution [Solved]



Solution

- Step 1. Mean $|r|$: 10.6 bp in the first bar and 5.1 bp at midday; ratio 2.06, 95% CI [1.98, 2.15].
- Step 2. Last 30 minutes: 18.6% of daily volume, 95% CI [18.3, 19.0]%, against $6/78 = 7.7\%$ if volume were even.
- Interpretation: bars of one day share its volatility and news, so resampling days keeps that dependence; days are treated as independent, although volatility persists; the narrow intervals show a systematic pattern, but the best execution time for a large buyer also depends on spreads, price impact, urgency and the risk of waiting.

B2: Spread Estimators at Two Frequencies [Solved]

Task (1/2)

□ Context and assumptions

- ▶ SPY 5-minute bars (`data/market/intraday/SPY.US_5m.csv`), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Daily SPY bars from `data/market/SPY.US.csv` (prices adjusted for dividends), September 2024 – September 2026. One tick is USD 0.01.

B2: Spread Estimators at Two Frequencies [Solved]

Task (2/2)

□ Sub-tasks

- ▶ (a) From the 5-minute bars of each day compute the Roll autocovariance, the mean Abdi–Ranaldo product and the mean two-bar Corwin–Schultz estimate (negatives set to 0; no overnight adjustment within a session). Average each over days; for Roll and Abdi–Ranaldo take the square root of the averaged moment (0 if its sign is wrong). Give 95% day-bootstrap intervals of the same statistics (1 000 resamples) Roll (1984), Corwin & Schultz (2012), Abdi & Ranaldo (2017).
- ▶ (b) Compute the same three estimators from the daily bars over the whole window (Corwin–Schultz with the overnight adjustment), with a moving-block bootstrap (blocks of 20 days, 1 000 resamples).
- ▶ (c) Compare every estimate with one tick relative to the price: a floor for the quoted spread, not for the effective spread.
- ▶ (d) Simulate the Roll model with SPY's bar volatility and 78 prices a day (1 477 days per sample, 200 samples), for a one-tick spread and for $c = 0$; compute the share of days with $\hat{\gamma}_1 > 0$ and the pooled estimate Harris (1990).

□ Report and interpret

- ▶ A table of estimates with intervals, and the Monte Carlo results next to the observed ones.
- ▶ Interpretation: which frequency gives a credible spread; what do the daily estimators measure; is the share of positive covariances evidence against the Roll model?

B2: solution [Solved]

Solution (1/2)

- Step 1 (a). 5-minute bars: Roll 2.82 bp [2.09, 3.43], Corwin–Schultz 3.11 bp [3.03, 3.20], Abdi–Ranaldo 1.06 bp [0.00, 1.67].
- Step 2 (b). Daily bars: Roll 64.3 bp [0.0, 114.4] (covariance ≥ 0 in 10% of resamples), Corwin–Schultz 26.5 bp [21.3, 33.7], Abdi–Ranaldo 55.1 bp [0.0, 92.2].
- Step 3 (c). One tick is 0.21 bp: even the 5-minute Corwin–Schultz estimate is 15 times larger; the daily estimators are two orders of magnitude above one tick, implausible as a cost for SPY, because the daily standard deviation is 103.4 bp.
- Step 4. Abdi–Ranaldo and daily Roll are square roots of averaged moments that can take the wrong sign in a resample (set to 0), so their intervals touch 0; the 5-minute Roll covariance is negative on only 59% of days.

B2: solution [Solved]

Solution (2/2)

- Step 5 (d). One-tick spread ($c = 0.10$ bp, bar s.d. 9.2 bp): $\hat{\gamma}_1 > 0$ on 45.3% of days [43.0, 48.1]; pooled Roll 2.07 bp [1.51, 2.53].
- Step 6 (d). No spread ($c = 0$): 45.3% [43.1, 48.3] and 2.06 bp [1.50, 2.53]: indistinguishable from a one-tick spread.
- Step 7 (d). Observed: 40.8% positive and 2.82 bp; bias-corrected Roll 1.92 bp.
- Interpretation: only intraday bars give credible magnitudes; daily estimators are dominated by volatility. Positive sample covariances can occur under Roll, but the observed share lies below the simulated band for this calibration; at $c/\sigma \approx 0.01$ the simulation barely distinguishes zero from a one-tick spread, and the 5-minute Roll estimate is mostly demeaning bias. The daily and intraday windows differ, so the frequency contrast also reflects the sample period.

B3: Spread Estimates Across 25 Assets [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Daily open, high, low, close and volume, September 2024 – September 2026, from data/market: 7 US stocks and ETFs, 11 BVB stocks and the BET ETF group, 7 crypto-assets (25 assets).
- ▶ Volatility annualised with 252 days for stocks and 365 for crypto; the Amihud ratio as in the lecture (bp per USD 1 million).

□ Sub-tasks

- ▶ (a) Compute Corwin–Schultz and Abdi–Ranaldo for each asset from daily bars, and the median of each group.
- ▶ (b) Compute the Spearman rank correlations of Corwin–Schultz with annualised volatility, of the Amihud ratio with the median traded value, and of Corwin–Schultz with Amihud.
- ▶ (c) Count the assets for which Abdi–Ranaldo is set to 0.

□ Report and interpret

- ▶ Group medians, the three correlations and the scatter plots.
- ▶ Interpretation: do BVB stocks look more or less liquid than US stocks; which measure would you trust to compare markets; what would you need to validate it?

□ Model: B2

B4: Illiquidity and the VIX [Solved]

Task

□ Context and assumptions

- ▶ SPY 5-minute bars (data/market/intraday/SPY.US_5m.csv), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Daily intraday illiquidity $ILLIQ_d$: the mean over the 78 bars of $|r|$ divided by the traded value, in bp per USD 100 million; VIX daily close from data/market/VIX.INDX.csv; 1 477 common days.

□ Sub-tasks

- ▶ (a) Estimate by OLS $\ln ILLIQ_d = a + b \ln VIX_d + e_d$.
- ▶ (b) Report OLS and HAC (Newey–West, 20 lags) standard errors, the 95% HAC interval, and the first-order autocorrelation of the residuals [Newey & West \(1987\)](#).
- ▶ (c) Add a linear time trend (in years) and re-estimate b with HAC errors.
- ▶ (d) Test the stability of b : with $D_d = 1$ from 2023-09-27 (the middle of the sample, fixed before estimation), estimate $\ln ILLIQ_d = a + b \ln VIX_d + gD_d + hD_d \ln VIX_d + e_d$ and test $h = 0$ with a HAC Wald test at the 5% level.

□ Report and interpret

- ▶ $\hat{b}$ with both standard errors, the trend regression and the break test.
- ▶ Interpretation: read b as an elasticity; why do the standard errors differ; is the elasticity stable; is the relation causal?

B4: solution [Solved]

Solution

- ▣ Step 1 (a). $\hat{b} = 0.95$, $R^2 = 0.30$: a 10% higher VIX goes with about $0.95 \times 10\%$ higher illiquidity, nearly proportional.
- ▣ Step 2 (b). OLS SE 0.038, HAC SE 0.121 (3.2 times larger); 95% HAC CI [0.72, 1.19]; residual autocorrelation 0.76.
- ▣ Step 3 (c). With a trend: $\hat{b} = 0.80$ (SE 0.120), trend -0.062 per year (SE 0.021): illiquidity falls over time, but the VIX link remains.
- ▣ Step 4 (d). Break on 2023-09-27: $b = 1.03$ before and $b + h = 0.60$ after; $\hat{h} = -0.43$ (HAC SE 0.25), Wald $p = 0.089 > 0.05$: stability is not rejected at 5%.
- ▣ Interpretation: persistent residuals make OLS errors far too small; the elasticity is close to 1 and weaker, but not significantly, in the second half; the relation is not causal, since both respond to news and uncertainty.

B5: Amihud Illiquidity on Three Markets [Proposed]

Solution discussed in class

Task

- ▣ Context and assumptions
 - ▶ The 25 assets of B3; daily Amihud ratio $|r_d|/DVOL_d$ in bp per USD 1 million, averaged over September 2024 – September 2026 [Amihud \(2002\)](#).
 - ▶ BVB traded value converted to USD at the BNR reference rate of each day.
- ▣ Sub-tasks
 - ▶ (a) Compute the median Amihud ratio of each group and the ratio BVB median / US median.
 - ▶ (b) Build a 95% bootstrap interval for that ratio by resampling assets within each group (1 000 resamples).
 - ▶ (c) Compute the Spearman correlation between the asset rankings of the first and the second year.
- ▣ Report and interpret
 - ▶ The medians, the ratio with its interval and the rank correlation.
 - ▶ Interpretation: how much less liquid is the BVB, how precise is that number, and is illiquidity a stable characteristic?
- ▣ **Model:** B2, B4

B6: Volume and Price Moves in SPY Bars [Solved]

Task (1/2)

□ Context and assumptions

- ▶ SPY 5-minute bars (data/market/intraday/SPY.US_5m.csv), regular session 09:30–16:00 New York time, only full days with 78 bars: 1 477 days, October 2020 – September 2026.
- ▶ Relative volume = bar volume / median volume at the same time of day; $z = |r|$ divided by the day's standard deviation of 5-minute returns; 115 052 bars.

B6: Volume and Price Moves in SPY Bars [Solved]

Task (2/2)

□ Sub-tasks

- ▶ (a) Sort bars into 20 equal-size groups by relative volume, regress $\ln \bar{z}$ on $\ln$ mean relative volume, and give a 95% day-bootstrap interval for the slope (300 resamples).
- ▶ (b) At bar level, regress $\ln(|r| + 10^{-6}) - \ln \sigma_{\text{day}}$ on $\ln$ relative volume with standard errors clustered by day, and test slope = 0.5 at 5%; the constant (0.01 bp) keeps bars with $r = 0$, where $\ln |r|$ is undefined: report their share and the slope without them.
- ▶ (c) IV: instrument $\ln$ relative volume by the log relative volume of the same bar on the previous day; report the first stage, the two-stage least squares slope and its clustered SE.

□ Report and interpret

- ▶ The grouped and bar-level slopes, the t statistic against 0.5, and the IV results.
- ▶ Interpretation: is the relation concave, is it the square-root law [Tóth et al. \(2011\)](#), why cluster by day, and is the exclusion restriction of (c) plausible?

B6: solution [Solved]

Solution

- ▣ Step 1 (a). Grouped slope 0.274, 95% CI [0.262, 0.289].
- ▣ Step 2 (b). Bar-level slope 0.309 (clustered SE 0.008); $t = (0.5 - 0.309)/0.008 \approx 24$, so slope = 0.5 is rejected at 5%; 0.8% of bars have $r = 0$, and without them the slope is 0.291 (SE 0.008).
- ▣ Step 3 (c). First stage slope 0.339 ($t = 32.8$): a relevant instrument; IV slope 0.063 (SE 0.018) against 0.309 by OLS.
- ▣ Interpretation: the relation is concave but flatter than a square root; a bar mixes buyers and sellers, while the law concerns one trader's net metaorder. The OLS–IV difference cannot establish how much of the OLS slope reflects simultaneity, because the exclusion restriction is doubtful: yesterday's volume at that time reflects persistent activity and scheduled news that also move today's $|r|$ (lag correlation of normalised $\ln|r|$: 0.043). Bars of one day share news, hence clustering.

B7: The Bitcoin Clock [Proposed]

Solution discussed in class

Task

- ▣ Context and assumptions
 - ▶ Bitcoin 5-minute bars (`data/market/intraday/BTC-USD.CC_5m.csv`), UTC, September 2024 – September 2026: 749 days, trading 24 hours a day, 7 days a week.
- ▣ Sub-tasks
 - ▶ (a) On weekdays, compute the mean $|r|$ in 13:00–17:00 UTC (US morning) and in 03:00–11:00 UTC (Asian and European morning), and their ratio with a 95% day-bootstrap interval (1 000 resamples).
 - ▶ (b) Compute the ratio of weekday to weekend mean $|r|$ with a 95% day-bootstrap interval.
- ▣ Report and interpret
 - ▶ The two ratios with intervals.
 - ▶ Interpretation: why does a 24/7 market have a clock, and when would you execute a large Bitcoin order?
- ▣ **Model:** B1

B8: Price Discovery, the BET ETF and the BET-TR Index [Solved]

Task (1/2)

□ Context and assumptions

- ▶ Daily log closes of the Patria-TV BET ETF (data/market/TVBETETF.R0.csv, days with trades) and of BET-TR (data/market/BETTR.INDX.csv), common days, September 2024 – September 2026 (lecture, Section 10).
- ▶ The ETF reinvests dividends, so it should be cointegrated with BET-TR with vector $\beta = (1, -1)'$; BET-TR is computed from its constituents' prices, not traded.
- ▶ Specification: an unrestricted constant in the Johansen test and in the VECM (it absorbs the equilibrium level of the ETF–index basis); lags of Δy chosen by BIC over 1–10.

B8: Price Discovery, the BET ETF and the BET-TR Index [Solved]

Task (2/2)

□ Sub-tasks

- ▶ (a) Run the Johansen trace test [Johansen \(1991\)](#), then estimate the VECM with $\beta = (1, -1)'$ imposed and lags chosen by BIC; test $\alpha_{\text{ETF}} = 0$ and $\alpha_{\text{idx}} = 0$ at 5%.
- ▶ (b) Compute the Hasbrouck information-share bounds [Hasbrouck \(1995\)](#) and the Gonzalo–Granger component share [Gonzalo & Granger \(1995\)](#), with 95% bootstrap intervals that resample moving blocks of 20 consecutive rows $(\Delta y_t, z_{t-1}, \Delta y_{t-1}, \dots)$ of the VECM regression, stationary quantities (1 000 resamples).
- ▶ (c) Re-estimate both α on each half of the sample.

□ Report and interpret

- ▶ The adjustment coefficients with t statistics and intervals, the share bounds, and the two half-samples.
- ▶ Interpretation: which price leads and how sure are you; why are the bounds so wide at daily frequency; why can the component share leave $[0, 1]$?

B8: solution [Solved]

Solution (1/2)

- Step 1 (a). BIC selects 1 lag; Johansen trace 49.6 for $r = 0$ (5% critical value 15.49) and 0.10 for $r \leq 1$ (3.84): one cointegrating vector; VECM with 493 observations: $\alpha_{\text{ETF}} = -0.12$ ($t = -2.11$, rejected at 5%), $\alpha_{\text{idx}} = -0.03$ ($t = -0.63$, not rejected).
- Step 2 (b). Bootstrap intervals: $\alpha_{\text{ETF}} [-0.29, 0.01]$, $\alpha_{\text{idx}} [-0.18, 0.09]$.
- Step 3 (b). Index information share in $[0.25, 0.98]$; bootstrap CI of the lower bound $[0.00, 0.64]$, of the upper $[0.72, 1.00]$; component share 1.39, CI $[-0.14, 3.32]$.

B8: solution [Solved]

Solution (2/2)

- Step 4 (c). First half: $\alpha_{\text{ETF}} = -0.26$ ($t = -2.34$); second half: -0.09 ($t = -0.85$); the index adjustment is not significant in either half ($t = -0.77$ and 0.47).
- Interpretation: the evidence suggests that the ETF adjusts toward the index (the index's α is not distinguishable from 0), but it is weak: the bootstrap interval for α_{ETF} touches 0 and the adjustment vanishes in the second half.
- Interpretation: the residual correlation of 0.93 at daily frequency makes the Cholesky ordering decisive, hence the wide bounds; both α being negative pushes the component share outside $[0, 1]$.

C1: Does Illiquidity Carry a Premium on the BVB?

Task (1/2)

□ Context and assumptions

- ▶ Open question and seed for the team project: how illiquid is the BVB compared with US and crypto markets, and is illiquidity priced?
- ▶ Amihud (2002): expected illiquidity should **raise** future returns; an unexpected rise should **lower** current returns [Amihud \(2002\)](#).
- ▶ Data: monthly BET-TR, SPY and Bitcoin returns from data/market; market illiquidity from the stocks (tokens) of each market, 2016–2026.

C1: Does Illiquidity Carry a Premium on the BVB?

Task (2/2)

□ Sub-tasks

- ▶ (a) Build a monthly market illiquidity index (cross-sectional mean of log Amihud, months with at least 10 trading days) and its shock, the residual of an AR(1).
- ▶ (b) Regress monthly returns on lagged illiquidity and the shock, with HAC standard errors (6 lags), and correct the lagged coefficient for the Stambaugh bias [Stambaugh \(1999\)](#) with the reduced-bias estimator [Amihud & Hurvich \(2004\)](#).
- ▶ (c) On the BVB, sort the ten blue chips each January by their Amihud ratio of the previous calendar year (stocks with more than 150 trading days); from monthly simple total returns form equal-weighted illiquid and liquid halves (weights reset monthly) and test the mean illiquid-minus-liquid return with HAC errors (6 lags).

□ Report and interpret

- ▶ A table per market, the scatter of returns on shocks, and a discussion of power: few stocks, ten years [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#).
- ▶ Interpretation: which of Amihud's two predictions survives on each market?

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have daily closes P (lei) and volumes V (number of shares, positive every day) for a BVB stock. Estimate the Roll spread and the Amihud illiquidity. What does a round trip cost me?"

The AI answer: text

- Roll: $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -c^2$ and $s = 2c$ Roll (1984)
- If the covariance comes out positive, take its absolute value: the formula still gives the spread
- Amihud: the average of $|r_d|$ divided by the volume of day d , per 1 million Amihud (2002)
- Result: $s = 0.05$ lei on a 20-lei stock, i.e. 0.25% of the price; a round trip (buy, then sell) costs $2s$, i.e. 0.50%

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np
dp = P.diff().dropna().values
g = np.cov(dp[1:], dp[:-1])[0, 1]
s_roll = 2 * np.sqrt(abs(g))
s_pct = 100 * s_roll / P.mean()
r = np.log(P).diff().dropna()
illiq = 1e6 * (r.abs() / V.loc[r.index]).mean()
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- (a) The answer contains three errors; everything else is correct. Identify each error in the text and, where applicable, in the code.
- (b) For each error, state how you would detect it: a test, a published number or a formula from the lecture.
- (c) Correct each error: give the formula, the code line where applicable, and a corrected number where the supplied information permits one.
- (d) Log each error as in `AI_ERRORS.md`; interpretation: which error would change an investment decision?
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- What sign of the autocovariance does the Roll model predict? What does the opposite sign tell you?
- Compare with the definition of $ILLIQ$ on the Formulas frame: what is $DVOL_d$?
- Draw the bid, the ask and the mid-price: at which prices do you buy and sell?

C3: The Price of the Speed Race in SPY

Task (1/2)

□ Context and assumptions

- ▶ Project seed from the lecture case study: [Aquilina, Budish & O'Neill \(2022\)](#) use exchange message data to measure the sniping race modelled by [Budish, Cramton & Shim \(2015\)](#): the latency arbitrage tax, daily race profits divided by daily traded value, is 0.419 bp (Table X).
- ▶ Their extrapolation models (Table XV, profits Π_t and traded value V_t in the same currency, coefficients in bp, 1 bp = 10^{-4}): col. 2, $\Pi_t = 0.4213 \cdot 10^{-4} V_t$; col. 6, $\Pi_t = 10^{-4}(0.3354 V_t + 0.0066 \sigma_t \bar{V})$, with σ_t the annualised realised volatility in percent and $\bar{V}$ the mean daily traded value; across the sensitivity scenarios of Table XIV the tax ranges from 0.20 to 0.74 bp.
- ▶ Open question: if the London tax applied to SPY, how large would the prize be, and how much would it move on stress days?
- ▶ Data: SPY 5-minute bars from data/market, regular US session (09:30–16:00 New York time), days with all 78 bars, the full years 2021–2025.

C3: The Price of the Speed Race in SPY

Task (2/2)

▣ Sub-tasks

- ▶ (a) For each day compute V_t , the sum of close times volume over the 78 bars, and $\sigma_t = 100\sqrt{252 \sum r^2}$ from the 5-minute log returns (the first bar from the open); report the yearly means of V_t and σ_t and the yearly correlation of V_t and σ_t with a 95% percentile bootstrap confidence interval that resamples days (1 000 resamples; days are treated as independent, so repeat with moving blocks of 20 days), and compare it with the 0.82 of the paper.
- ▶ (b) With $\bar{V}$ equal to the mean of the year, compute for each year the implied annual prize $\sum_t \Pi_t$ under col. 2, col. 6 and the lowest and highest scenarios of Table XIV.
- ▶ (c) Show that under col. 6 the annual tax $\sum_t \Pi_t / \sum_t V_t$ equals $0.3354 + 0.0066 \bar{\sigma}$ bp, with $\bar{\sigma}$ the mean volatility of the year, and find the $\bar{\sigma}$ above which col. 6 gives a larger prize than col. 2.
- ▶ (d) Plot the daily tax at mean traded value ($V_t = \bar{V}$), $0.3354 + 0.0066 \sigma_t$, over 2021–2025, list the five highest days, and explain why the realised ratio $\Pi_t / V_t = 0.3354 + 0.0066 \sigma_t \bar{V} / V_t$ differs.

▣ Report and interpret

- ▶ The table of (a) with the intervals, the prize table of (b), the derivation and threshold of (c), and the plot and list of (d).
- ▶ Interpretation: which assumptions carry a tax measured in London in 2015 over to SPY today, and in which direction could the fragmentation of US trading and the use of 5-minute instead of 1-minute volatility move your estimate? Can the data at hand sign these effects?

Common Pitfalls

- ▣ Reading a daily high-low spread estimate as the true spread of a liquid asset
- ▣ Comparing illiquidity across markets without converting the traded value to one currency
- ▣ Treating 5-minute bars of the same day as independent observations
- ▣ Mixing price and volume adjustment bases across a stock split, or computing returns from unadjusted prices
- ▣ Calling a volume–volatility relation in bars a price-impact law

Take-Home Assignment (Team Project)

- Choose three assets not covered in class (a BVB stock outside the ten blue chips, a crypto token, a US small cap)
- Compute Amihud and the two high-low spread estimators by month; plot them with the volatility
- Estimate the cost of selling USD 1 million within one day with the square-root law: $Y = 1$ (and 0.5, 1.5 as sensitivity), σ the daily return volatility, Q/V the position over the median daily traded value; report the peak impact and the average shortfall ($\frac{2}{3}$ of it), in bp and in USD
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

Further Study on Quantinar

Advanced courses on quantinar.com

-  Applied Computational Finance

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_10

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